



DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
COMPUTER ENGINEERING
THIRD SESSIONAL

SUBJECT: (CE-313) DATABASE MANAGEMENT SYSTEMS

Examination : B.Tech Semester - III
Date : 11/Oct/2022
Time : 09:00 a.m. to 10:15 a.m.

Seat No. :
Day : Tuesday
Max. Marks : 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

- Q.1 Do as directed.** [12]
- CO3 A** (a) Examine: "Starvation is possible when validation based protocols are used". [2]
- CO3 N** (b) Explain the need of lock compatibility matrix in brief. [2]
- CO3 R** (c) Define: IS lock and IX lock. [2]
- CO4 U** (d) What is the role of autonomous transactions? [2]
- CO1 A** (e) Create a procedure that takes two numbers and returns their minimum. [2]
- CO4 R** (f) Explain ACID properties in brief. [2]
- Q.2** Attempt **Any TWO** from the following questions. *finish* [12]
- CO3 E** (a) Consider the given statement: "Unlocking may occur earlier in graph based protocols as compared to lock based protocols". Justify with a suitable example and explanation of graph based protocol. [6]
- CO5 E** (b) Explain and compare following deadlock prevention schemes with examples: [6]
1. Wait-Die Scheme
 2. Wound-Wait Scheme
- CO3 E** (c) Justify: "Thomas write rule can avoid rollback in case of obsolete write operations." Support your answer with a detailed explanation of the modified version of time stamp ordering protocol which includes thomas write rule. [6]
- Q.3** Attempt the following questions. [12]
- CO5 N** (a) Test whether the following schedule is conflict serializable or not. [6]
- $S = r_4(A), r_2(A), r_3(A), w_1(B), w_2(A), r_3(B), w_2(B)$
Here, r_i and w_i denote read and write operations respectively on some data item in transaction i .
Also find all the possible serialized schedules if and only if the conflict serializability test passes.
- CO3 C** (b) Perform Extendible Hashing on the following elements: [6]
1. 16, 20, 7, 27, 29, 18, 11, 30, 23
Bucket Size: 2
Hash Function: Suppose the global depth is X . Then the Hash Function returns X LSBs.
- OR**
- Q.3** Attempt the following questions. [12]
- CO4 N** (a) Explain the recovery actions, when the database system is restarted after a crash. [4]
- Mention the log records added during recovery for following logged actions. [2]
- $\langle T_0 \text{ start} \rangle$
 $\langle T_0, B, 2000, 2050 \rangle$
 $\langle T_1 \text{ start} \rangle$
 $\langle \text{checkpoint } \{T_0, T_1\} \rangle$
 $\langle T_1, C, 700, 600 \rangle$
 $\langle T_1 \text{ commit} \rangle$
 $\langle T_2 \text{ start} \rangle$
 $\langle T_2, A, 500, 400 \rangle$
 $\langle T_0, B, 2000 \rangle$
CRASH
- CO4 R** (b) Explain the state diagram of a transaction. [3]
- CO4 U** (c) Justify: The presence of a checkpoint record in the log allows the system to streamline its recovery procedure. [3]

$$\text{WTS}(0) = 2$$

$$\text{RTS}(0) = 1$$

Re

$$1 < 0$$

$$2 < 1$$

$$2 < 0$$

$$1 < 1$$

$$1 < 2$$

ignored

$$\begin{array}{|c|} \hline 2 \\ \hline 16, 20 \\ \hline \end{array}$$

00

$$\begin{array}{|c|} \hline 2 \\ \hline 1, 29 \\ \hline \end{array}$$

01

$$\begin{array}{|c|} \hline 2 \\ \hline 18, 30 \\ \hline \end{array}$$

10

$$\begin{array}{|c|} \hline 3 \\ \hline 27, 11 \\ \hline \end{array}$$

011

$$\begin{array}{|c|} \hline 3 \\ \hline 7, 23 \\ \hline \end{array}$$

111

Tot

1342

1432

4132

